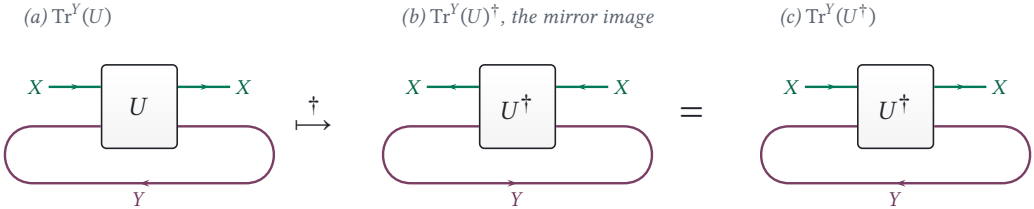
(a) $\mathrm{Tr}^Y(U)$
(b) $\mathrm{Tr}^Y(U)^\dagger$, the mirror image
(c) $\mathrm{Tr}^Y(U^\dagger)$
X
U
X
Y
†
X
$U^\dagger$
X
Y
=
X
$U^\dagger$
X
Y